

Program : Diploma in Polymer Technology	
Course Code : 3093	Course Title: Technology Of Elastomers
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3,T:1,P:0)	Periods per semester: 60

Course Objectives:

- To impart the basic concepts of rubber compounding, principles of compounding and vulcanization.
- To understand the cure characteristics and different vulcanization methods
- To design various recipes to meet vulcanisate properties.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in polymer science.		Fundamentals of Polymer science	2
Basic knowledge in Engineering chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Generalise the classification, functions and properties of compounding ingredients.	15	Understanding
CO2	Explain the principles of rubber compounding and vulcanization.	10	Understanding
CO3	Determine the cure characteristics, cure measurements and different vulcanisation methods for elastomers.	15	Understanding
CO4	Apply principles of rubber compounding for the design of various recipes to meet the processing requirements and vulcanizate properties.	18	Applying

	Series Test	2	
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CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Generalise the classification, functions and properties of compounding ingredients.		
M1.01	Classify the compounding ingredients.	2	Remembering
M1.02	Identify the functions and characteristics of base polymer, vulcanizing agents, activators, retarders, accelerators, antidegradants, processing aids with examples.	5	Understanding
M1.03	Describe the classification, functions, properties of black and non-black fillers, production of thermal and furnace blacks.	5	Understanding
M1.04	List the special purpose additives with their functions and examples.	3	Understanding
Contents: Compounding ingredients: classification - base polymer, activators and retarders, accelerators, fillers, processing aid, antidegradants, vulcanizing agents, and special additives - examples, characteristics and their functions in a compound. Vulcanising agents - sulphur, sulphur donors, peroxides, metal oxides, resin. Carbon black - types - furnace, thermal, channel and lamp black - method of manufacture - furnace process and thermal process. Properties - particle size, structure, physical nature of surface, chemical nature of surface and particle porosity. Classification - ASTM D 1765. Procedure - particle size, structure. Mineral fillers – Precipitated Silica ,China clay, Precipitated CaCO ₃ ,whiting, Aluminium silicate,TiO ₂ . Organic fillers, Fibrous fillers -Functions ,Properties .Special purpose additives - functions-examples			
CO2	Explain the principles of rubber compounding and vulcanization.		
M2.01	State the principles of rubber compounding and its objectives.	3	Remembering

M2.02	Enlist the steps involved in formulating a recipe.	2	Understanding
M2.03	Explain mastication, masterbatch, gum & filled compound	2	Understanding
M2.04	Summarize curing systems and curing of elastomers.	3	Understanding
	Series Test – I	1	

Contents:

Principles of compounding and Vulcanisation: Compounding - definition - objectives - elements to be considered for formulating a recipe - importance of base polymer - recipe for rubber products. Mastication, master batch, gum and filled compound. Justify curing of elastomers - sulphur and non - sulphur curing systems - NR and SR-types of crosslinks - effects on vulcanizate properties. Sulphur curing systems - CV, EV, and Semi EV. Chemistry of sulphur curing, Non - sulphur curing system - CR, CSM, EPM .

CO3	Determine the cure characteristics ,cure measurements and different vulcanisation methods for elastomers		
M3.01	Recall cure characteristics, and explain the procedure to determine them using Mooney viscometer and rheometers.	3	Understanding
M3.02	Illustrate various cure graphs by changing different type of accelerators and curing agents.	3	Understanding
M3.03	Explain compression moulding, transfer moulding and injection moulding.	4	Understanding
M3.04	Describe batch curing and continuous curing methods.	5	Understanding

Contents:

Cure terms, Cure measurements and Vulcanisation methods:

Curing - define cure terms - scorch, scorch time, cure and cure time, optimum cure, state of cure, cure index. Scorch safety - methods to avoid scorch-

Cure graph - Mooney viscometer, ODR-MDR. Procedure to determine cure characteristics. Cure curves - by changing accelerators and curing agents..

Moulding – Definition,Types of moulding-compression, transfer and injection.

Batch curing - Autoclave - water curing - cold curing - hot air oven curing.

Continuous curing methods - steam tube curing - hot air tunnel - fluidized bed - molten salt - rotocure - microwave curing - examples.

CO4	Apply principles of rubber compounding for the design of various recipes to meet the processing requirements and vulcanizate properties		
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M4.01	Chart out the properties of elastomers and general principles of rubber compounding.	3	Understanding
M4.02	Identify the elements to be considered for designing a formulation for a product.	5	Understanding
M4.03	Design compounds for Particular hardness , specific gravity ,volume cost calculations.	5	Applying
M4.04	Generalize the factors to be considered for compounding to meet processing requirements and vulcanisate properties.	5	Understanding
	Series Test – II	1	

Contents:

Compounding for processing and vulcanisate properties

General principle of compounding - chart out properties of elastomer. Compounding to meet processing requirements - compounding to meet vulcanisate properties - hardness, tensile, tear resistance, abrasion - oil resistance, heat resistance, flame resistance - oxygen and ozone resistance - formulation. Calculation of specific gravity and volume cost. Design compounds for particular Shore A hardness using black and non-black fillers.

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd, 1982 Edition)
R1	Rubber Technology -Maurice Morton (Publisher – Springer, 1999 Edition)
R2	Rubber Technology Hand book - Werner Hoffman (Publisher – Oxford University Press,1989 Edition)
R3	Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)

Online Resources:

Sl.No	Website Link
1	Rubberchemtechnol,org
2	www.rubberatkins.com
3	https://www.science direct.com
4	https://www.utwente.nl